

MOORED BUOY 21004, Japan

WMO: 992280

Lat: 29.000N

Lon: 135.000E

Elev: 3

StdP: 14.69

Time Zone: 9.00 (E09)

Period: 90-00

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	53.3	54.8	35.8	30.8	55.9	37.8	33.3	56.7	43.9	58.9	40.7	57.9	20.8	340	0.556

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.7	86.0	79.7	85.4	79.4	84.9	79.3	81.8	84.1	81.2	83.9	80.7	83.8	10.7	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
81.0	161.3	83.4	80.5	158.1	83.2	79.8	154.8	83.1	45.7	84.1	45.0	84.0	44.4	83.7	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
46.8	36.2	30.6	50.5	87.7	1.3	3.0	49.6	89.9	48.8	91.6	48.1	93.3	47.1	95.5		
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	71.9	62.0	60.8	63.7	67.2	71.3	76.7	81.8	82.7	81.5	77.0	71.5	65.5
	DBStd	8.30	3.92	3.90	3.63	3.49	2.98	2.98	2.19	1.83	2.16	3.04	3.21	3.85
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	361	108	123	69	19	0	0	0	0	0	0	1	41
	CDD50	7986	372	304	424	515	659	801	987	1015	946	838	644	481
	CDD65	2874	16	7	28	84	195	351	522	550	496	373	195	57
	CDH74	23232	0	0	2	19	280	2199	5853	6516	5496	2556	305	6
	CDH80	5629	0	0	0	1	1	151	1653	2147	1464	211	1	0
Wind	WSAvg	15.0	16.8	17.2	15.0	13.6	12.8	13.9	13.4	14.0	16.2	16.5	16.1	15.4
Precipitation	PrecAvg	68.5	3.0	3.0	3.8	6.3	10.1	11.9	4.7	4.8	6.0	7.0	4.4	3.7
	PrecMax	100.4	8.0	7.9	7.8	13.7	20.0	24.8	11.4	12.3	14.1	14.1	8.8	8.2
	PrecMin	51.9	1.7	1.3	1.2	1.1	2.8	4.0	0.4	0.6	1.6	2.6	1.2	1.3
	PrecStd	8.9	1.5	1.5	1.7	2.9	4.2	5.2	3.1	2.9	3.2	3.0	1.9	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.8	70.6	72.6	76.1	80.1	83.3	87.0	86.7	86.2	83.6	79.1	74.9
		MCWB	67.4	67.2	69.4	74.4	77.7	79.1	79.8	79.9	80.8	77.7	74.6	70.5
	2%	DB	70.2	69.1	71.0	73.9	78.8	82.4	86.1	85.9	85.4	82.8	77.4	73.5
		MCWB	65.8	65.3	68.1	71.9	75.3	78.4	79.5	79.7	79.7	77.2	72.6	68.2
	5%	DB	69.1	67.8	70.0	72.5	77.3	81.7	85.3	85.4	84.9	82.1	76.6	72.5
		MCWB	64.2	63.3	67.3	70.6	74.1	78.0	79.2	79.5	79.4	76.8	71.3	66.6
	10%	DB	67.9	66.6	69.0	71.5	75.3	80.9	84.6	85.0	84.4	81.3	75.8	71.2
		MCWB	62.2	61.4	66.1	69.5	72.2	77.6	78.9	79.4	79.1	76.2	70.1	64.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.1	68.7	70.1	72.7	75.2	80.9	82.4	82.9	82.2	80.0	76.3	72.4
		MCDB	71.0	70.3	71.0	72.9	75.8	81.8	84.4	84.3	84.4	82.0	77.8	74.2
	2%	WB	67.2	66.2	68.7	71.3	73.9	80.0	81.6	81.9	81.1	78.9	74.6	69.8
		MCDB	69.8	68.4	69.8	71.8	74.7	81.2	84.1	84.0	84.2	81.6	76.7	72.8
	5%	WB	65.1	64.3	67.8	70.3	73.1	79.2	80.9	81.4	80.6	78.1	72.8	67.6
		MCDB	68.6	67.3	69.0	71.1	74.1	80.6	83.9	84.0	84.0	81.2	75.8	71.7
	10%	WB	62.9	62.1	66.5	69.4	72.1	78.4	80.4	80.8	80.0	77.4	71.4	65.8
		MCDB	67.5	66.0	68.3	70.4	73.5	79.9	83.7	83.9	83.7	80.7	75.0	70.6
Mean Daily Temperature Range	5% DB	MDBR	4.5	4.4	4.3	3.7	3.2	2.9	3.0	2.7	3.2	3.4	3.9	4.1
		MCDBR	6.0	5.2	5.0	4.1	3.5	2.6	3.6	2.9	3.3	3.4	4.5	4.7
		MCWBR	8.2	8.0	6.2	4.5	5.7	2.0	2.2	2.0	2.5	3.0	5.2	6.3
	5% WB	MCDBR	5.9	5.4	4.6	3.7	3.5	2.9	3.5	2.4	3.3	3.1	4.4	4.6
MCWBR		8.8	8.7	6.1	5.1	4.2	2.6	2.9	2.2	2.8	3.4	6.0	7.1	
Clear-Sky Solar Irradiance	taub	0.427	0.455	0.505	0.525	0.498	0.463	0.465	0.466	0.459	0.459	0.433	0.426	
	taud	2.196	2.136	2.009	2.008	2.179	2.365	2.357	2.355	2.351	2.286	2.295	2.206	
	Ebn at Noon	251	255	250	250	256	264	263	262	261	252	249	245	
	Edh at Noon	39	45	55	56	48	39	40	39	38	39	36	37	
All-Sky Solar Radiation	RadAvg	825	1022	1338	1554	1597	1633	2077	1995	1694	1275	962	765	
	RadStd	76	101	106	148	198	195	115	81	103	86	80	62	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A